

13 Cardinality of Sets

13.1 Sets with Equal Cardinalities

A. Show that the two given sets have equal cardinality by describing a bijection from one to the other. Describe your bijection with a formula (not as a table).

2. $\mathbb{R}$ and $(\sqrt{2}, \infty)$

$$f(x) = e^x + \sqrt{2}, \text{ with inverse } f^{-1}(y) = \ln(y - \sqrt{2}).$$

4. The set of even integers and the set of odd integers

$$f(n) = n + 1, \text{ with inverse } f^{-1}(m) = m - 1.$$

6. $\mathbb{N}$ and $S = \{\frac{\sqrt{2}}{n} : n \in \mathbb{N}\}$

$$f(n) = \frac{\sqrt{2}}{n}. \text{ It is surjective by the definition of } S, \text{ and injective since } \frac{\sqrt{2}}{n} = \frac{\sqrt{2}}{m} \text{ gives } n = m.$$

8. $\mathbb{Z}$ and $S = \{x \in \mathbb{R} : \sin x = 1\}$

$$\text{Since } \sin x = 1 \text{ exactly when } x = \frac{\pi}{2} + 2\pi k \text{ for some } k \in \mathbb{Z}, \text{ take } f(k) = \frac{\pi}{2} + 2\pi k, \text{ with inverse } f^{-1}(x) = \frac{x - \frac{\pi}{2}}{2\pi}.$$

10. $\{0, 1\} \times \mathbb{N}$ and $\mathbb{Z}$

$$f(a, b) = a - 2ab + b, \text{ shown bijective in Exercise 12 of Section 12.2.}$$

12. $\mathbb{N}$ and $\mathbb{Z}$ (Suggestion: use Exercise 18 of Section 12.2.)

$$f(n) = \frac{(-1)^n(2n-1)+1}{4}, \text{ shown bijective in Exercise 18 of Section 12.2.}$$

14. $\mathbb{N} \times \mathbb{N}$ and $\{(n, m) \in \mathbb{N} \times \mathbb{N} : n \leq m\}$

$$f(n, m) = (n, n + m - 1), \text{ with inverse } f^{-1}(a, b) = (a, b - a + 1).$$

B. Answer the following questions concerning bijections from this section.

16. Verify that the function f in Example 13.3 is a bijection.

$$\text{Here } f : (0, \infty) \rightarrow (0, 1) \text{ is given by } f(x) = \frac{x}{x+1}.$$

It is injective: $\frac{x}{x+1} = \frac{y}{y+1}$ gives $x(y+1) = y(x+1)$, that is, $xy + x = xy + y$, so $x = y$.

It is surjective: given $y \in (0, 1)$, the number $x = \frac{y}{1-y}$ lies in $(0, \infty)$ and satisfies $f(x) = y$.

13.2 Countable and Uncountable Sets

2. Prove that the set $A = \{(m, n) \in \mathbb{N} \times \mathbb{N} : m \leq n\}$ is countably infinite.

Since $\mathbb{N}$ is countably infinite, Theorem 13.5 gives that $\mathbb{N} \times \mathbb{N}$ is countably infinite. By Exercise 14 of Section 13.1 the map $f(m, n) = (m, m + n - 1)$ is a bijection $\mathbb{N} \times \mathbb{N} \rightarrow A$, so $|\mathbb{N} \times \mathbb{N}| = |A|$. Therefore $|\mathbb{N}| = |A|$ and A is countably infinite.

4. Prove that the set of all irrational numbers is uncountable. (Suggestion: Consider proof by contradiction using Theorems 13.4 and 13.6.)

Proof. Let $I = \mathbb{R} - \mathbb{Q}$ be the set of irrational numbers. First note that I is infinite, since the numbers $\sqrt{2} + k$ for $k \in \mathbb{N}$ are distinct and all irrational.

Suppose for the sake of contradiction that I is not uncountable. Being infinite, I must then be countably infinite. By Theorem 13.4 the set $\mathbb{Q}$ is countably infinite, so Theorem 13.6 gives that $\mathbb{Q} \cup I$ is countably infinite. But $\mathbb{Q} \cup I = \mathbb{R}$, so $|\mathbb{N}| = |\mathbb{R}|$, contradicting Theorem 13.2.

Therefore I is uncountable. □

6. Prove or disprove: There exists a bijective function $f : \mathbb{Q} \rightarrow \mathbb{R}$.

This is false. By Theorem 13.4 the set $\mathbb{Q}$ is countably infinite, so $|\mathbb{N}| = |\mathbb{Q}|$. If a bijection $f : \mathbb{Q} \rightarrow \mathbb{R}$ existed we would have $|\mathbb{Q}| = |\mathbb{R}|$, and hence $|\mathbb{N}| = |\mathbb{R}|$, contradicting Theorem 13.2. So no such bijection exists.

8. Prove or disprove: The set $\mathbb{Z} \times \mathbb{Q}$ is countably infinite.

This is true. By Exercise 12 of Section 13.1 the set $\mathbb{Z}$ is countably infinite, and by Theorem 13.4 so is $\mathbb{Q}$. Theorem 13.5 then gives that $\mathbb{Z} \times \mathbb{Q}$ is countably infinite.

10. Prove or disprove: The set $A = \{\frac{\sqrt{2}}{n} : n \in \mathbb{N}\}$ is countably infinite.

This is true. By Exercise 6 of Section 13.1 the map $f(n) = \frac{\sqrt{2}}{n}$ is a bijection $\mathbb{N} \rightarrow A$, so $|\mathbb{N}| = |A|$ and A is countably infinite.

12. Describe a partition of $\mathbb{N}$ that divides $\mathbb{N}$ into $\aleph_0$ countably infinite subsets.

For each $k \in \mathbb{N}$ let $X_k = \{2^{k-1}(2m-1) : m \in \mathbb{N}\}$ the set of naturals divisible by exactly $k-1$ factors of 2. The first few are

$$X_1 = \{1, 3, 5, 7, 9, \dots\},$$

$$X_2 = \{2, 6, 10, 14, 18, \dots\},$$

$$X_3 = \{4, 12, 20, 28, 36, \dots\}.$$

Then $P = \{X_k : k \in \mathbb{N}\}$ is the required partition. Every $n \in \mathbb{N}$ can be written as $n = 2^a(2m-1)$ for a unique pair of integers $a \geq 0$ and $m \geq 1$, obtained by dividing out every factor of 2, so n lies in exactly one X_k , namely $k = a + 1$. Each X_k is countably infinite, since $m \mapsto 2^{k-1}(2m-1)$ is a bijection $\mathbb{N} \rightarrow X_k$. Finally there is one such set for each $k \in \mathbb{N}$, and these are distinct, so P has $\aleph_0$ elements.

14. Suppose $A = \{(m, n) \in \mathbb{N} \times \mathbb{R} : n = \pi m\}$. Is it true that $|\mathbb{N}| = |A|$?

Yes. The condition $n = \pi m$ determines n from m , so $A = \{(m, \pi m) : m \in \mathbb{N}\}$. The map $f : \mathbb{N} \rightarrow A$ given by $f(m) = (m, \pi m)$ is surjective by this description, and injective because $f(m) = f(m')$ forces $m = m'$ in the first coordinate. Hence $|\mathbb{N}| = |A|$.

13.3 Comparing Cardinalities

2. Prove that the set $\mathbb{C}$ of complex numbers is uncountable.

Proof. Every real number x is the complex number $x + 0i$, so $\mathbb{R} \subseteq \mathbb{C}$. The set $\mathbb{R}$ is infinite, and $|\mathbb{N}| \neq |\mathbb{R}|$ by Theorem 13.2, so $\mathbb{R}$ is uncountable. By Theorem 13.9, any set containing an uncountable subset is uncountable, so $\mathbb{C}$ is uncountable. $\square$

4. Prove or disprove: If $A \subseteq B \subseteq C$ and A and C are countably infinite, then B is countably infinite.

This is true. Since A is countably infinite it is infinite, and $A \subseteq B$, so B is infinite. Thus B is an infinite subset of the countably infinite set C , and Theorem 13.8 gives that B is countably infinite.

6. Prove or disprove: Every infinite set is a subset of a countably infinite set.

This is false. For a counterexample take $\mathbb{R}$, which is infinite. Suppose $\mathbb{R} \subseteq S$ for some countably infinite set S . Then $\mathbb{R}$ is an infinite subset of S , so Theorem 13.8 would make $\mathbb{R}$ countably infinite, contradicting Theorem 13.2. So $\mathbb{R}$ is not a subset of any countably infinite set.

8. Prove or disprove: The set $\{(a_1, a_2, a_3, \dots) : a_i \in \mathbb{Z}\}$ of infinite sequences of integers is countably infinite.

This is false; the set is uncountable. Write S for this set, and note S is infinite, as it contains the constant sequence $(k, k, k, \dots)$ for every $k \in \mathbb{Z}$.

Suppose for the sake of contradiction that S is countably infinite. By Theorem 13.3 its elements can be arranged in an infinite list $s_1, s_2, s_3, \dots$, where we write

$$s_k = (a_{k1}, a_{k2}, a_{k3}, \dots).$$

Now define a new sequence $b = (b_1, b_2, b_3, \dots)$ by setting $b_i = a_{ii} + 1$. Each b_i is an integer, so $b \in S$. But for every k the sequences b and s_k differ in their k th entry, since $b_k = a_{kk} + 1 \neq a_{kk}$. So b does not appear in the list, contradicting the assumption that the list contains every element of S .

Therefore S is not countably infinite, and being infinite it is uncountable.

10. Prove that if A and B are finite sets with $|A| = |B|$, then any surjection $f : A \rightarrow B$ is also an injection. Show this is not necessarily true if A and B are not finite.

Proof. Let $|A| = |B| = n$ and suppose $f : A \rightarrow B$ is surjective. Suppose for the sake of contradiction that f is not injective, so there are elements $a \neq a'$ in A with $f(a) = f(a')$. The set $f(A)$ has one element for each distinct value of f , and since a and a' contribute the same value, the n elements of A produce at most $n - 1$ distinct values. Hence $|f(A)| \leq n - 1$. But f is surjective, so $f(A) = B$ and $|f(A)| = n$. This gives $n \leq n - 1$, a contradiction. Therefore f is injective. $\square$

This can fail when the sets are infinite. Take $A = B = \mathbb{N}$ and define $f : \mathbb{N} \rightarrow \mathbb{N}$ by

$$f(n) = \begin{cases} 1 & \text{if } n = 1, \\ n - 1 & \text{if } n \geq 2. \end{cases}$$

Then f is surjective, since any $m \in \mathbb{N}$ equals $f(m+1)$. But f is not injective, because $f(1) = f(2) = 1$.

13.4 The Cantor-Bernstein-Schröder Theorem

2. Show that $|\mathbb{R}^2| = |\mathbb{R}|$. Suggestion: Begin by showing $|(0, 1) \times (0, 1)| = |(0, 1)|$.

Proof. By the Cantor-Bernstein-Schröder theorem it is enough to give an injection each way between $(0, 1) \times (0, 1)$ and $(0, 1)$.

One way, $x \mapsto (x, \frac{1}{2})$ is clearly injective.

The other way, give each $x \in (0, 1)$ its unique decimal expansion $0.a_1a_2a_3\dots$ that does not end in all 9's, and interleave digits:

$$(x, y) \mapsto 0.a_1b_1a_2b_2a_3b_3\dots$$

This expansion does not end in all 9's either, since that would force both x and y to, so its digits can be read off and separated back into those of x and y . Hence the map is injective, and $|(0, 1) \times (0, 1)| = |(0, 1)|$.

By Example 13.4 there is a bijection $h : \mathbb{R} \rightarrow (0, 1)$, so $(x, y) \mapsto (h(x), h(y))$ is a bijection $\mathbb{R}^2 \rightarrow (0, 1) \times (0, 1)$. Therefore

$$|\mathbb{R}^2| = |(0, 1) \times (0, 1)| = |(0, 1)| = |\mathbb{R}|. \quad \square$$

4. Let F be the set of all functions $\mathbb{R} \rightarrow \{0, 1\}$. Show that $|\mathbb{R}| < |F|$.

Proof. The map $\Phi : F \rightarrow \mathcal{P}(\mathbb{R})$ given by $\Phi(f) = \{x \in \mathbb{R} : f(x) = 1\}$ is a bijection: it is injective because a function into $\{0, 1\}$ is determined by where it takes the value 1, and surjective because any $X \subseteq \mathbb{R}$ is $\Phi(f)$ for the function f that is 1 on X and 0 elsewhere. So $|F| = |\mathcal{P}(\mathbb{R})|$.

By Theorem 13.7, $|\mathbb{R}| < |\mathcal{P}(\mathbb{R})|$, and transporting this across Φ gives $|\mathbb{R}| < |F|$. $\square$

6. Show that $|\mathcal{P}(\mathbb{N} \times \mathbb{N})| = |\mathcal{P}(\mathbb{N})|$.

Proof. By Theorem 13.5 the set $\mathbb{N} \times \mathbb{N}$ is countably infinite, so there is a bijection $g : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$. Then

$$\Phi : \mathcal{P}(\mathbb{N} \times \mathbb{N}) \rightarrow \mathcal{P}(\mathbb{N}), \quad \Phi(X) = g(X)$$

is a bijection. It is injective because g is, so $g(X) = g(Y)$ gives $X = g^{-1}(g(X)) = g^{-1}(g(Y)) = Y$. It is surjective because g is, so any $Z \subseteq \mathbb{N}$ equals $g(g^{-1}(Z)) = \Phi(g^{-1}(Z))$. (Both steps use Exercise 12 of Section 12.6.)

Therefore $|\mathcal{P}(\mathbb{N} \times \mathbb{N})| = |\mathcal{P}(\mathbb{N})|$. $\square$